25901043

MAHATMA GANDHI UNIVERSITY, KOTTAYAM

MGU-BVOC (HONOURS) Regular EXAMINATION November 2025

First SEMESTER

Skill Development Components (SDC) -MG1SDCFTA102-MATHEMATICAL AND STATISTICAL METHODS IN FOOD TECHNOLOGY

(2025 ADMISSION ONWARDS)

Duration: 2 hours SCHEME FOR EVALUATION Maximum Marks: 7

Part A

One or two sentence

Answer any 5 questions

Each question carries 2 marks

1. Correct Calculation . ……..…… ..…1

   Answer : -2 ……,,……. 1

2. A nilpotent matrix is a square matrix ,,,,,,,,,,…….,1
   A^k=0 for some positive integer k … ……..……1
3. Apply the chain rule……1
   $3(xsinx)^{2(}$sinx+xcosx)……1
4. Apply the power rule…… ……1
   $42x^6$-$54x^5$ ……..…1
5. Each merits 1 mark…….……(1x2)….2
6. Correct definition……2
7. Correct definition……2
8. Any Crrect Example ……...2

Part B

Short Answer

Answer any 5 questions

Each question carries 6 marks

9. det(A) = 7……..2

adj(A) = $\begin{matrix} 3 & 1 \\ -1 & 2 \end{matrix}$ ……..……2

A^{-1} …… $\begin{matrix} 3/7 & 1/7 \\ -1/7 & 2/7 \end{matrix}$ ……..2

Total……(6)

10. Upper Triangular matrix with Example……3

Lower Triangular matrix with Example……3

Total……(6)

11. Expand the Expression(i) ……1

$X^4/4+x^3+x^2+C$ ……2

Expand the Expression(ii) ……1

$3\ln - 2/x + 3/2x^2$ ……2

Total……(6)

12. A good measure of dispersion should possess several key properties to be a reliable statistical tool

Simplicity and Ease of Calculation……1

Rigidly Define……..1

Based on All Observations……1

Amenable to Further Algebraic Treatment……..1

Sampling Stability……1

Unaffected by Extreme Values……1

Total……(6)

13. the formula for the arithmetic mean…… 2

Sum the given incomes $\sum x=522$, $n=10$……2

$\bar{x}=52.2$……2

Total……(6)

14. The Z-distribution, or standard normal distribution, is a specific normal distribution with a mean of zero ……2

Standardization……1

Z-score Formula: The Z-score for a value……1

Probability Calculation……**2**

Total……(6)

15. Find $P(Z \leq 1.8)$……2
$P(Z>1.8)=1-P(Z \leq 1.8)$……1
$P(Z>1.8)=1-0.9641$……1
$P(Z>1.8)=0.0359$……2

Total……(6)

Part C

Essay

Answer any 2 questions

Each question carries 15 marks

16. Det(A) = -5……3
AdjA……3
Formula of A^{-1} ……..3

Computation of A^{-1} ……3
Matrix AdjA……3
Total……(15)

17. Expand the expression……..5
Differentiate the polynomial……5
$18x^5+15x^4-20x^3+48x+42x-35$……5
Total……(15)

18. Cmpute cumulative frequencies & total frequency(N = 40)……5
Q3 = 26.5……..4
D2 =19.29……..3
P4 =21.94……3
Total……(15)
19. Any five properties (3 x5)……15